

WEEK 6 – POWER BI REPORT

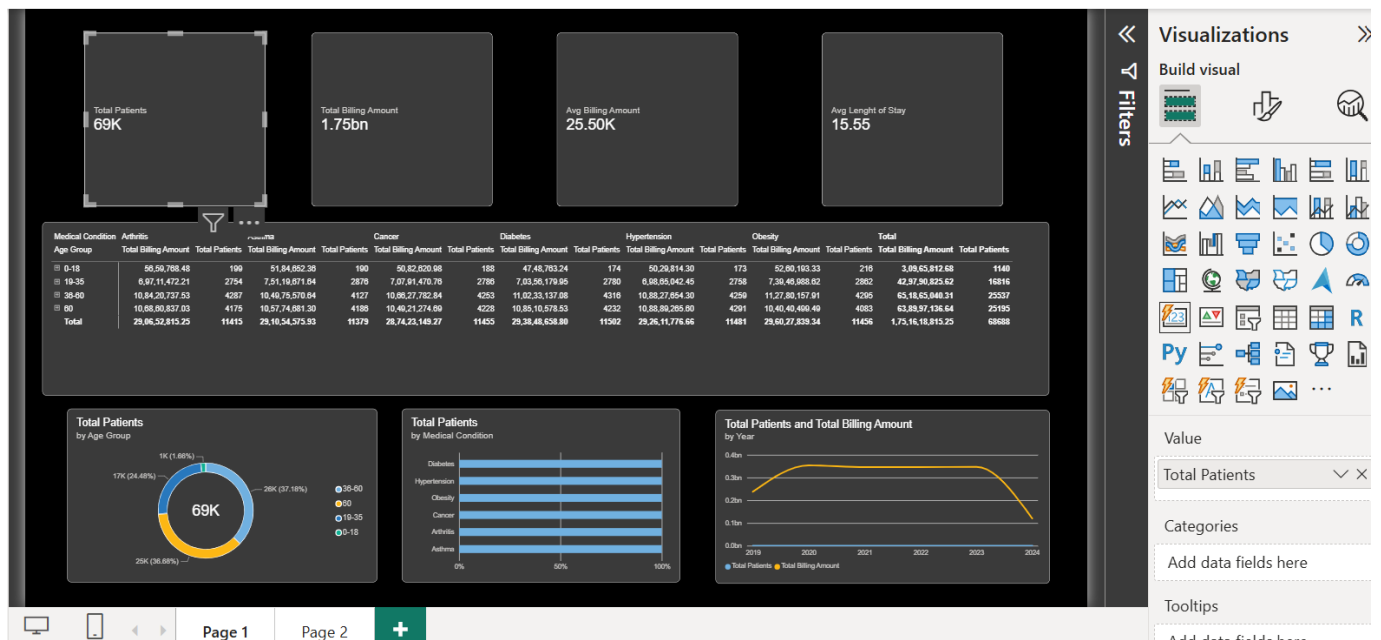
Healthcare Data Analysis Dashboard

This report documents the Power BI visuals created for healthcare data analysis, including KPI cards, matrix, charts and interactive slicers.

Tools Used: Microsoft Power BI

Dashboard Focus: Patient, Billing, Admission and Insurance Analysis

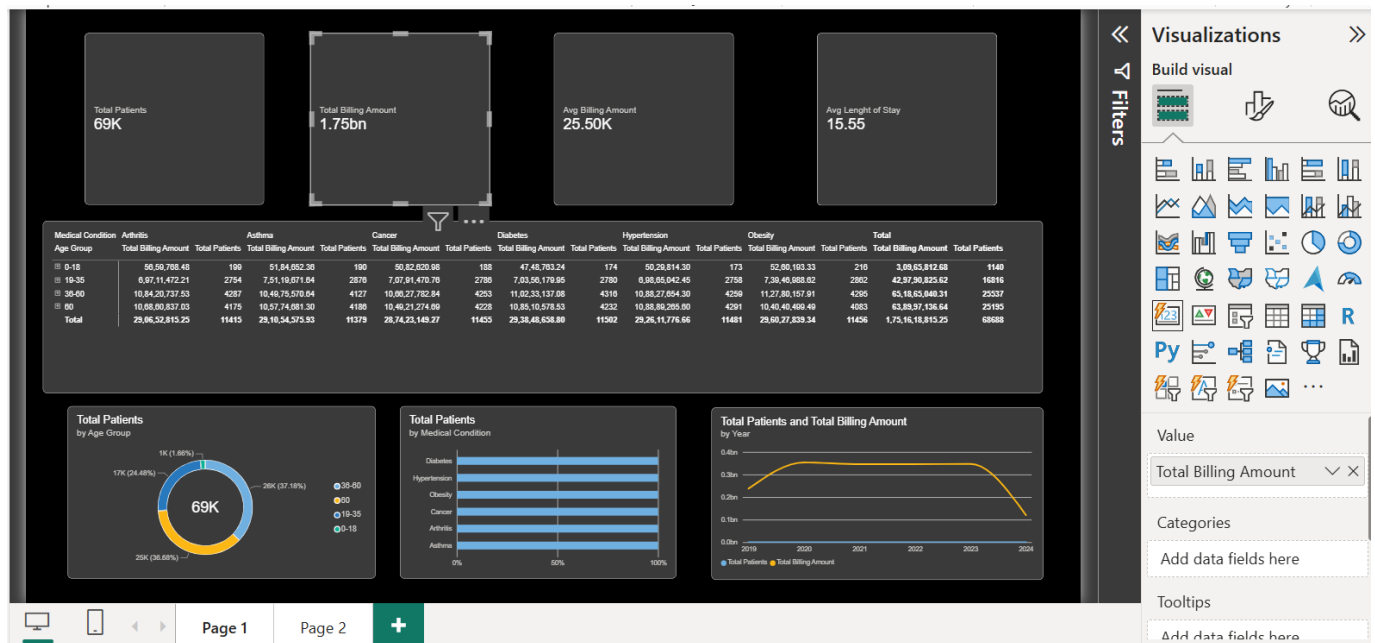
Screenshot 1 – Total Patients Card



Total Patients KPI Card

A Card visual was created to display the total number of patients in the healthcare dataset. The card shows a total of **69K patients**. This provides a quick overview of the overall patient count.

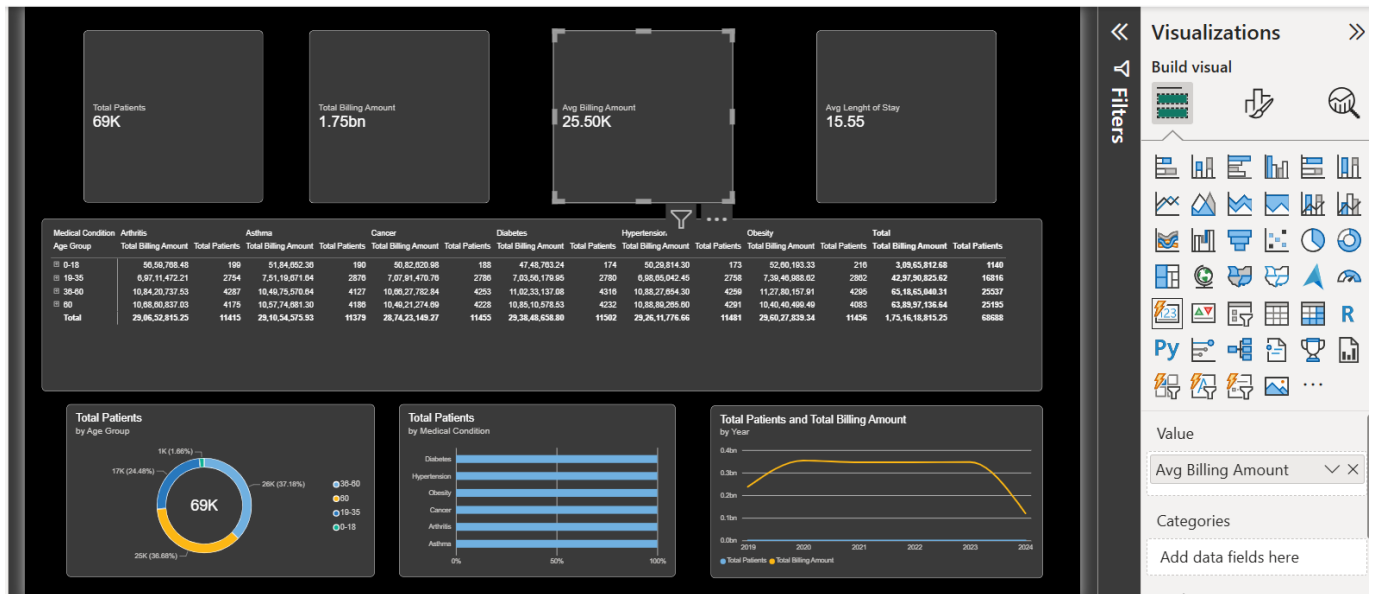
Screenshot 2 – Total Billing Amount Card



Total Billing Amount KPI Card

A Card visual was created to calculate the overall billing amount. The dashboard displays a total billing amount of approximately **1.75 billion**. This KPI helps to understand the total revenue generated from patient billing.

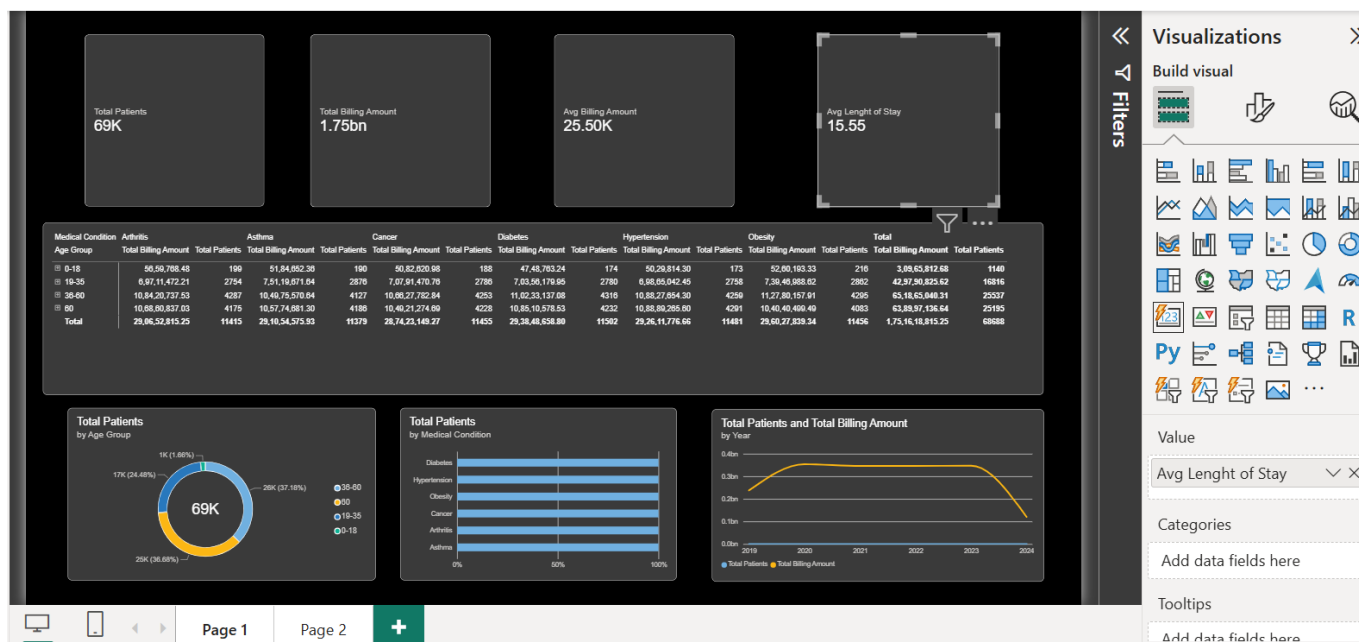
Screenshot 3 – Average Billing Amount Card



Average Billing Amount KPI Card

A Card visual was used to display the average billing amount per patient. The calculated average billing amount is approximately **25.50K**. This helps in understanding the average cost associated with patient treatment.

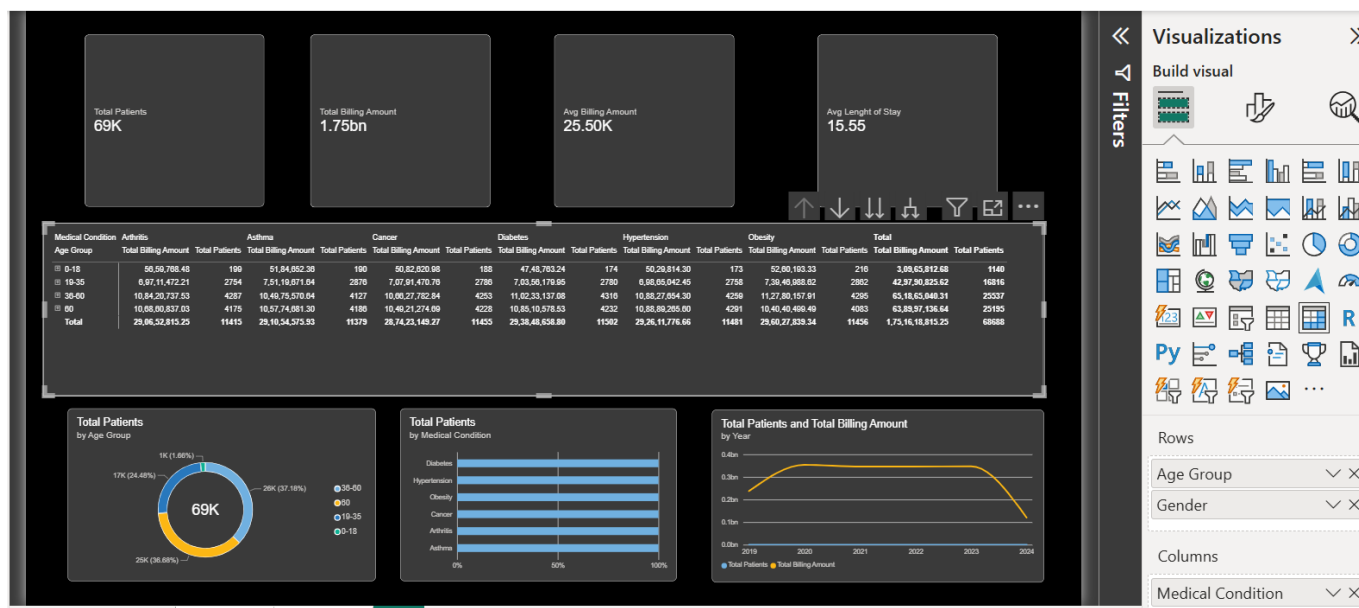
Screenshot 4 – Average Length of Stay Card



Average Length of Stay KPI Card

A Card visual was created to show the average number of days patients stayed in the hospital. The dashboard displays an average length of stay of **15.55 days**. This KPI helps analyse the typical duration of hospitalization.

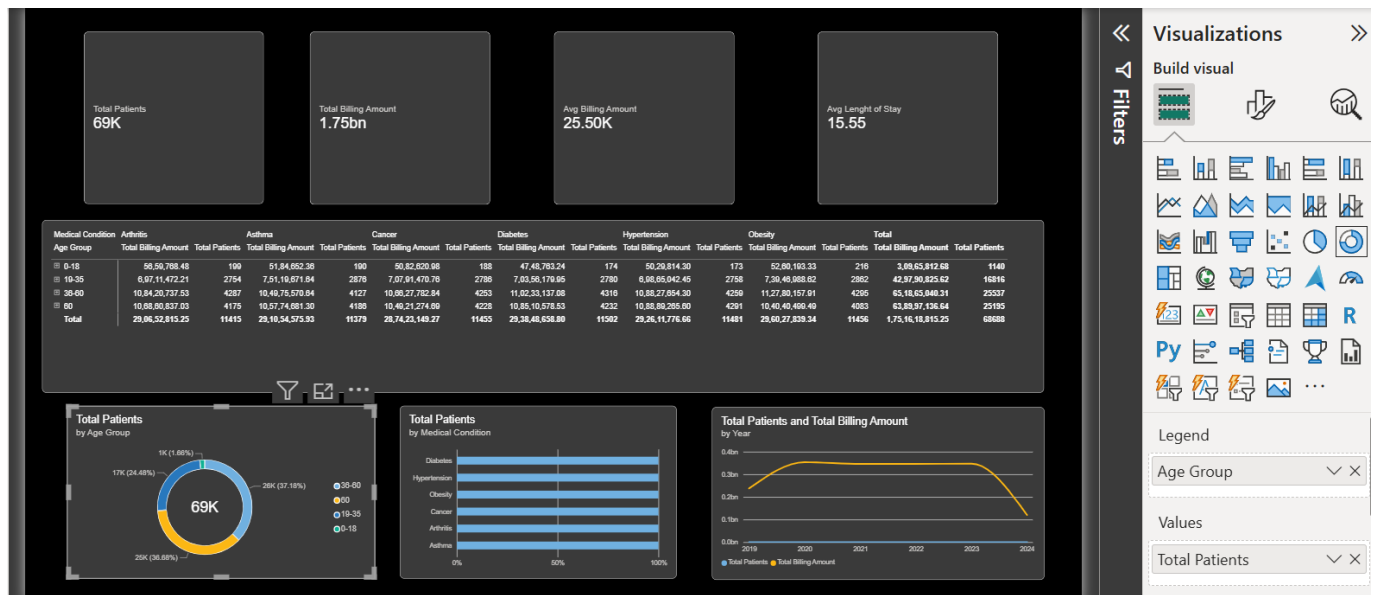
Screenshot 5 – Matrix Visual



Patient and Billing Analysis using Matrix

A Matrix visual was created to analyse patient and billing information based on **Age Group**, **Gender** and **Medical Condition**. The matrix displays **Total Billing Amount** and **Total Patients** for different medical conditions such as Arthritis, Asthma, Cancer, Diabetes, Hypertension and Obesity. This visual helps compare patient counts and billing amounts across different age groups and medical conditions.

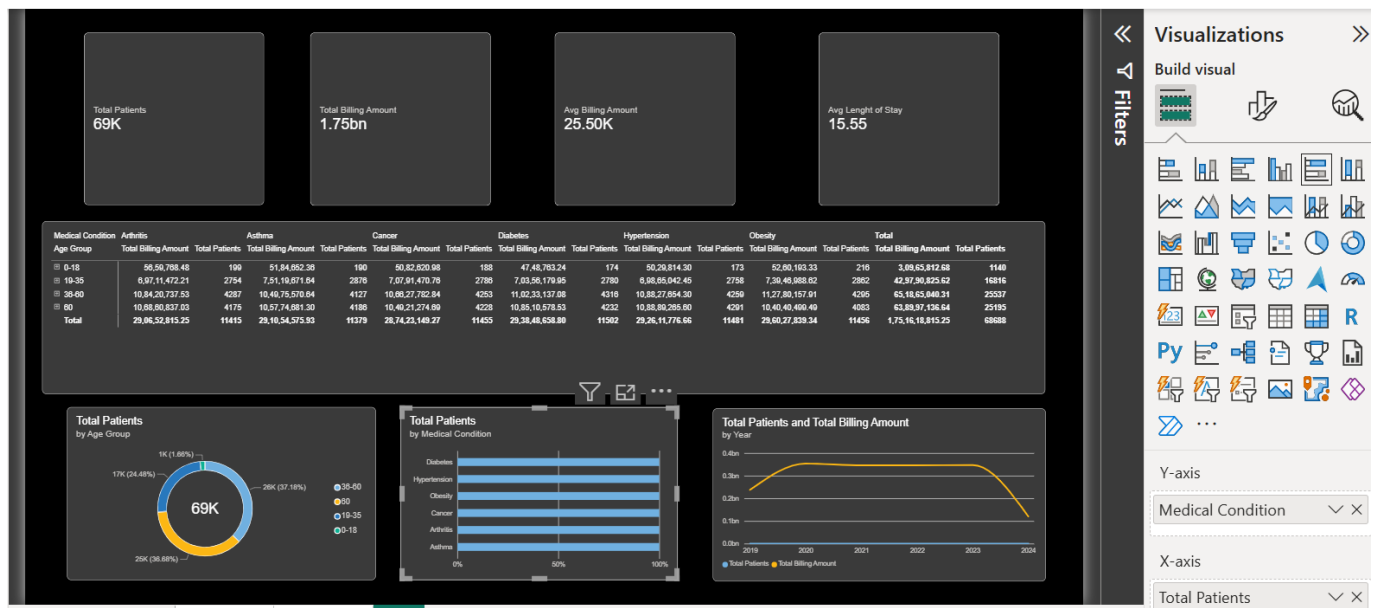
Screenshot 6 – Donut Chart



Total Patients by Age Group

A Donut Chart was created to represent the distribution of patients across different age groups. The age groups used are **0–18**, **19–35**, **36–60** and **60+**. The chart provides an easy visual comparison of the number and percentage of patients in each age group.

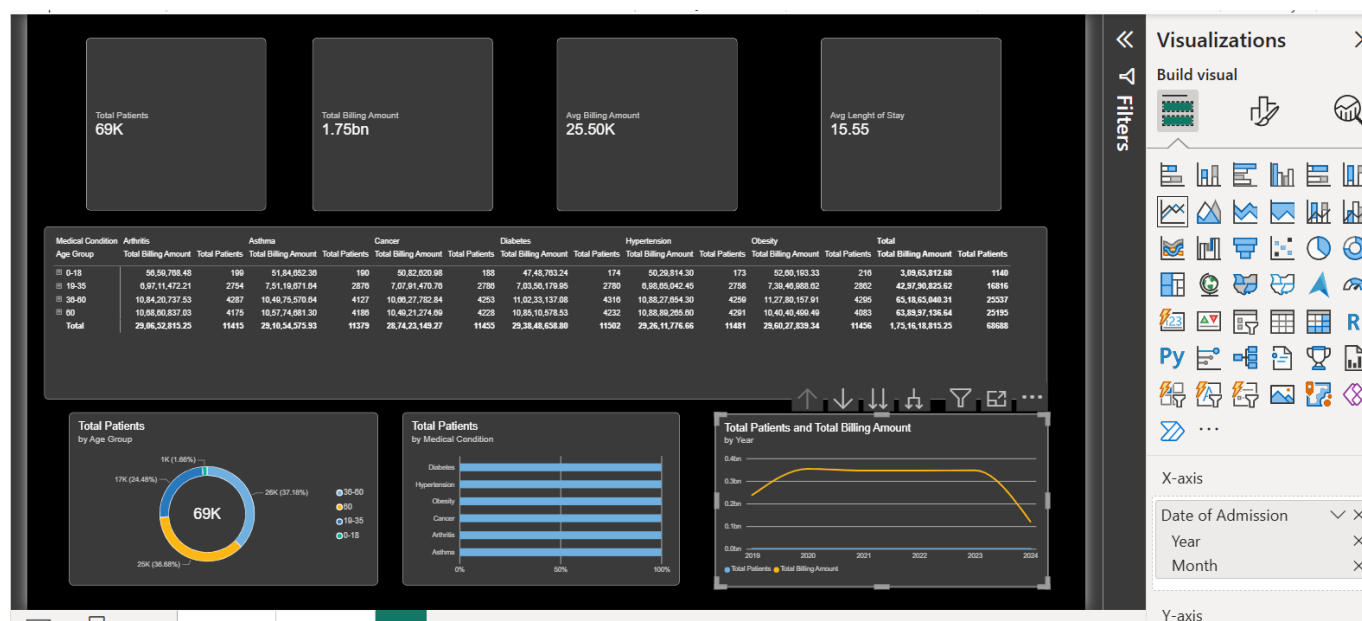
Screenshot 7 – Bar Chart



Total Patients by Medical Condition

A Bar Chart was created to analyse the total number of patients based on their medical condition. The chart includes conditions such as **Diabetes, Hypertension, Obesity, Cancer, Arthritis and Asthma**. This visualization helps compare the patient distribution among different medical conditions.

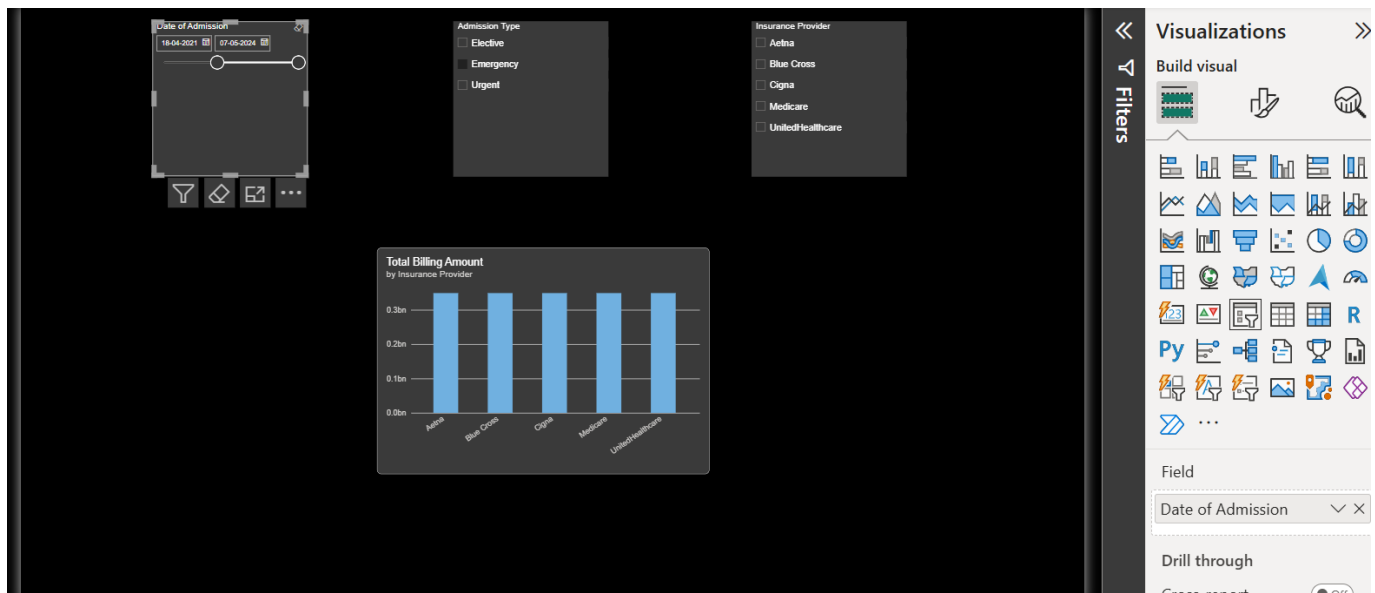
Screenshot 8 – Line Chart



Total Patients and Total Billing Amount by Year

A Line Chart was created to analyse the yearly trend of **Total Patients** and **Total Billing Amount**. The **Date of Admission** field was used for the time-based analysis. The chart helps identify changes and trends in patient admissions and billing amounts over different years.

Screenshot 9 – Date of Admission Slicer



Date of Admission Filter

A Date Slicer was added using the **Date of Admission** field. The slicer allows users to select a specific date range and filter the dashboard accordingly. This makes the dashboard interactive and helps users analyse healthcare information for a particular admission period.

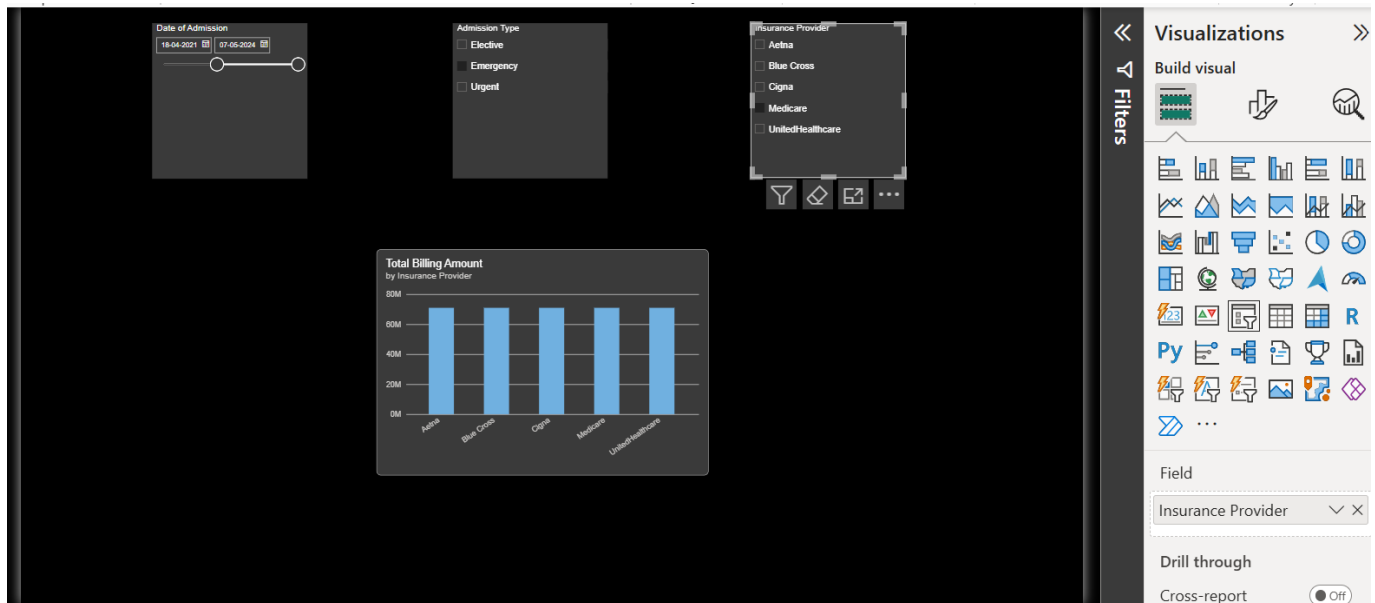
Screenshot 10 – Admission Type Slicer



Admission Type Filter

A Slicer was created using the **Admission Type** field. The available categories are **Elective**, **Emergency** and **Urgent**. Users can select an admission type to filter the dashboard and analyse the corresponding patient and billing information.

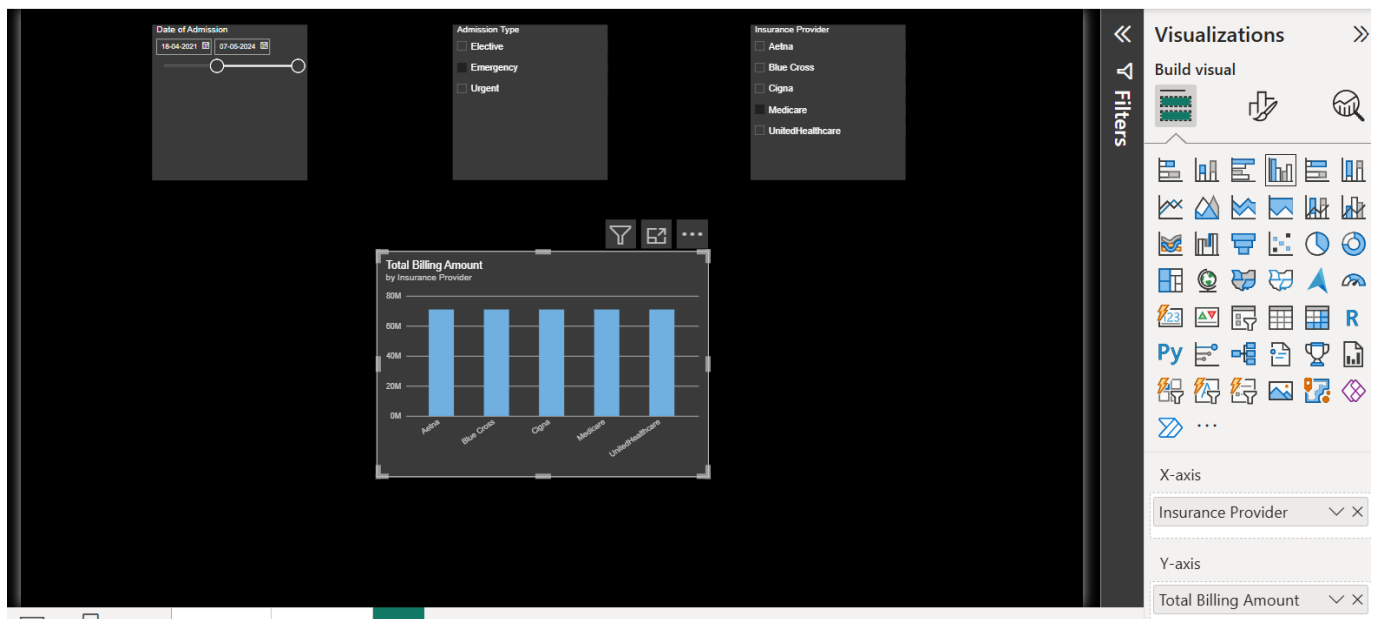
Screenshot 11 – Insurance Provider Slicer



Insurance Provider Filter

A Slicer was created using the **Insurance Provider** field. The available insurance providers include **Aetna, Blue Cross, Cigna, Medicare and UnitedHealthcare**. This slicer allows users to filter the dashboard based on the selected insurance provider.

Screenshot 12 – Total Billing Amount by Insurance Provider



Billing Analysis by Insurance Provider

A Column Chart was created to display the **Total Billing Amount by Insurance Provider**. Insurance Provider was used on the X-axis and Total Billing Amount was used on the Y-axis. The chart compares the billing amounts associated with **Aetna, Blue Cross, Cigna, Medicare and UnitedHealthcare**. It provides a clear comparison of billing performance across different insurance providers.

Overall Conclusion

The Week 6 Power BI task successfully demonstrates the creation of an **interactive healthcare analytics dashboard**. Different visualizations such as **KPI Cards, Matrix, Donut Chart, Bar Chart, Line Chart and Slicers** were used to analyse patient information, medical conditions, billing amounts, admission trends and insurance providers. The dashboard provides a clear and interactive way to understand the healthcare dataset and supports data-driven analysis.